

QUESTION 1. [Start a new page]

- (a) Calculate, correct to one decimal place

$$\frac{7.04}{8.04 - \sqrt{27.04}}$$

- (b) Mary bought a dress for \$75 during the sales. How much did she save if the sales discount on the dress was 40%?

- (c) If $f(x) = \frac{1}{1+x^3}$.

- (i) Evaluate $f(1)$ and $f(2)$.

- (ii) Using the trapezoidal rule with three function values, find the approximate area under the curve between $x=0$ and $x=2$.

- (d) Evaluate $\log_4 \left(\frac{1}{\sqrt{2}} \right)$.

- (e) Factorise $a^2 - b^2 + 2a + 2b$.

QUESTION 2. [Start a new page]

- (a) Solve the equation $\frac{1}{2}(x+1) - \frac{1}{5}(x-2) = 3$.

- (b) Solve:

(i) $x^2 + 6x - 16 = 0$.

(ii) $x^2 + 6x - 16 \geq 0$.

- (c) A box contains 4 red and 3 green apples. Peter took two apples at random.

- (i) Find the probability that they are:

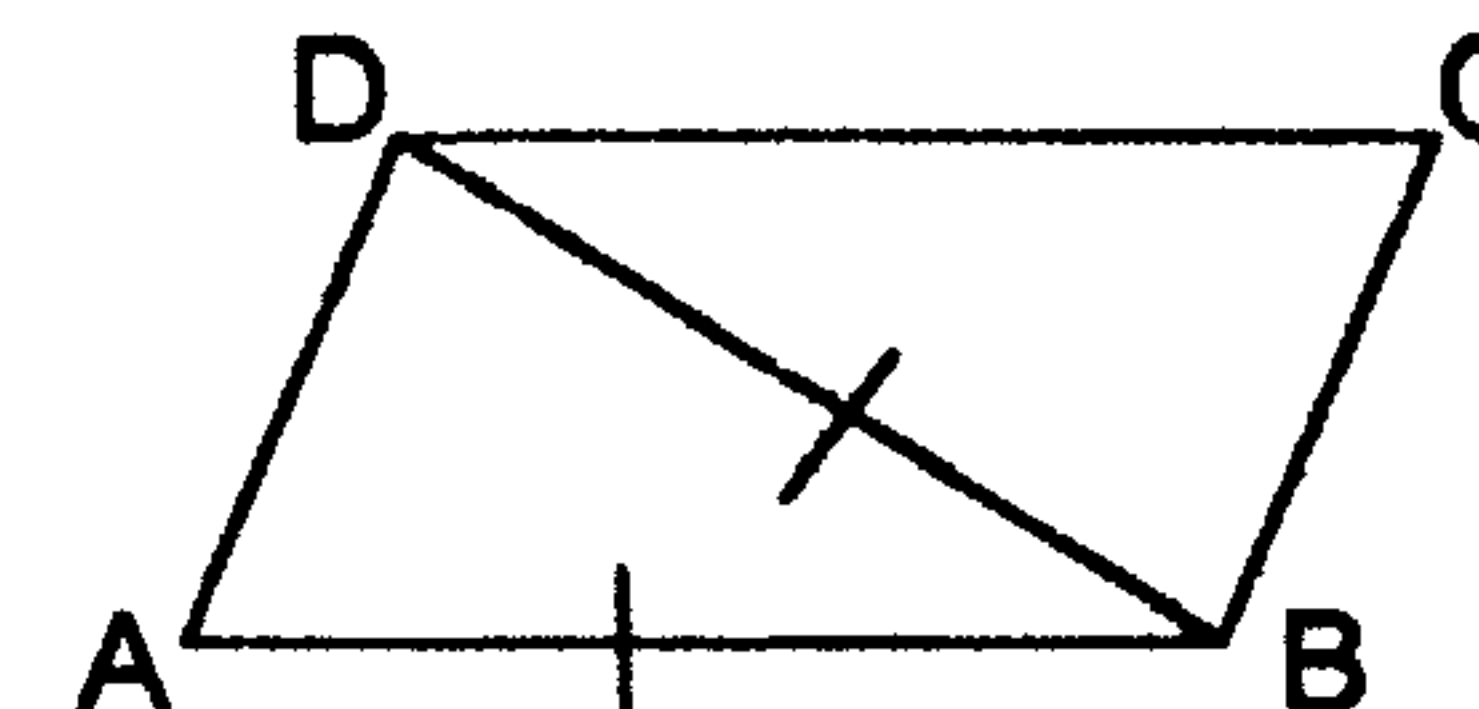
(1) both red.

(2) one of each colour.

- (ii) If it is known that at least one of the apples is red, find the probability that they are both red.

QUESTION 3. [Start a new page]

- (a)



ABCD is a parallelogram. Angle ADC equals 120° . Diagonal DB equals side AB .

- (i) Find $\angle DAB$ (give reasons).

- (ii) Show that triangle ABD is equilateral (give reasons).

- (iii) If smaller diagonal $DB = 8$ cm, find the area of the parallelogram.

- (b) Consider the parabola with equation $y = x^2 + 2x - 5$.

- (i) Express the equation in the form $y - k = (x - h)^2$, where a , h , and k are constants.

- (ii) Write down the coordinates of the vertex and the coordinates of the focus.

- (c) The roots of the equation $x^2 - 4x + 7 = 0$ are α and β . Evaluate $\frac{1}{\alpha} + \frac{1}{\beta}$.

QUESTION 4. [Start a new page]

A particle is moving in a straight line such that its displacement x from a fixed point O is given by:

$$x = 3t^2 - t^3, \quad x \text{ in metres, } t \text{ (in seconds)} \geq 0.$$

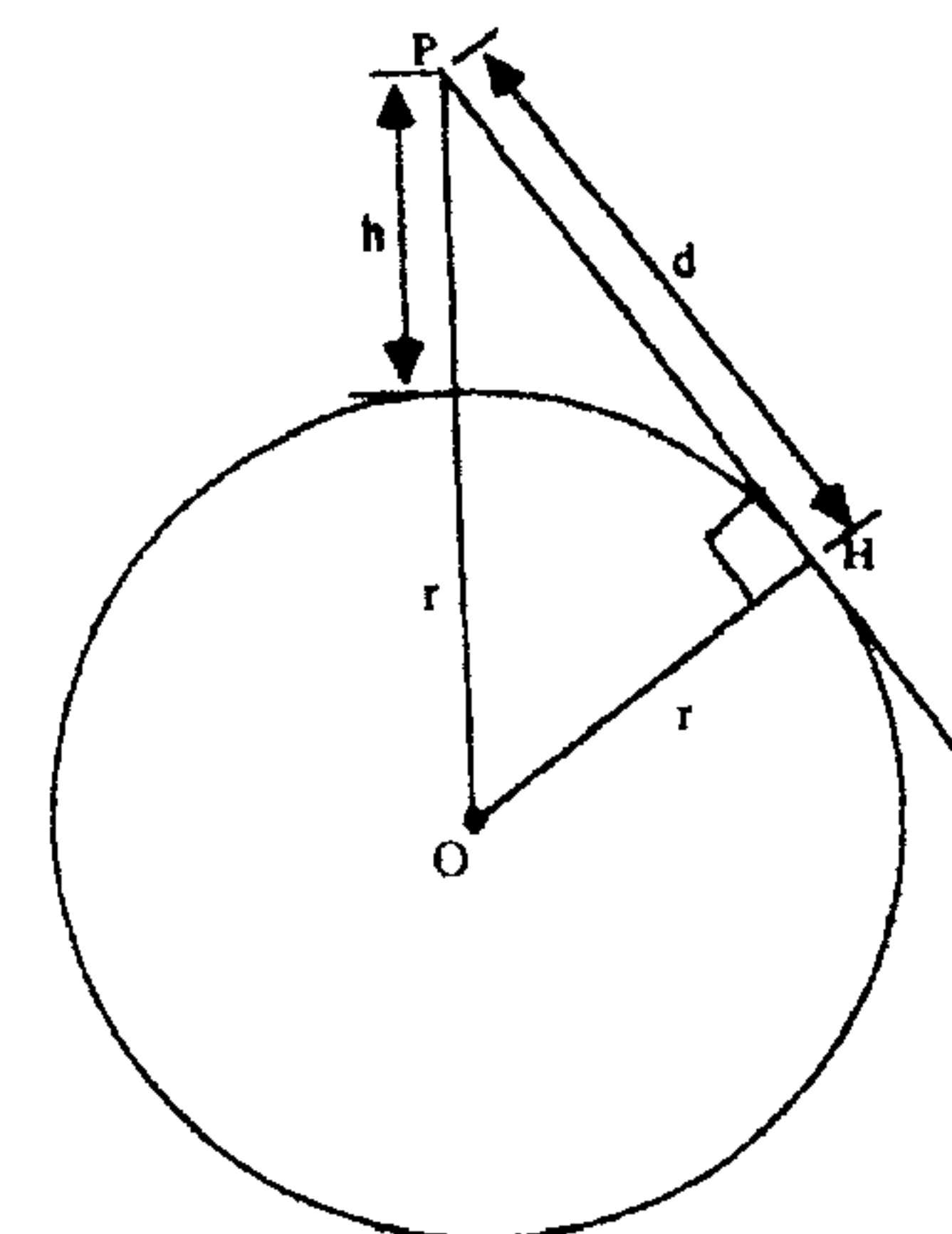
- Find the initial velocity and acceleration of the particle.
- Find the stationary points on the displacement-time curve and state their nature.
- Evaluate time at which the acceleration of the particle is zero.
- Graph the function $x(t)$, $t \geq 0$.
- Evaluate the total distance travelled by the particle in the first 4 seconds.
- Describe the motion, making use of the information obtained above.

QUESTION 5. [Start a new page]

- The first term of an arithmetic sequence is 1 and the fifteenth is 29.
 - Find the value of the common difference.
 - Find the sum of the first 150 terms.
- The equation of a parabola is $y = \frac{1}{4}x^2$ and the points $P(8, p)$ and $Q(q, 1)$ are on the curve.
 - If P and Q are on opposite sides of the axis of symmetry of the parabola, find p and q .
 - Show that the mid point M of segment PQ is $(3, 8\frac{1}{2})$.
 - The equation of the tangent to the parabola at Q is $y = -x - 1$. Show that the equation of the tangent at P is $y = 4x - 16$.
 - Find the coordinates of the point T , the intersection of the two tangents.
 - Show that the parabola bisects the segment TM .

QUESTION 6. [Start a new page]

- Find the area of a sector of 20° cut from a circle of radius 5 cm.
- The diagram (not to scale) represents the planet Mars (spherical shape assumed) and an astronaut at P , h metres above the surface. PH is the distance d , in metres, to the horizon, and r is the radius of Mars, also in metres.



- Show that d and h are connected by the relation $h^2 + 2hr - d^2 = 0$.
- Hence show that $h = r\sqrt{1 + \frac{d^2}{r^2}} - r$.
- When k is very small, say $k < 10^{-4}$, then $\sqrt{1+k}$ can be approximated by $1 + \frac{1}{2}k$. Show that in such a case, $h = \frac{d^2}{2r}$ approximately.
- If the observer at P has his horizon 20 km distant and the radius of Mars is 3398 km,
 - Show that the formula in (iii), $h = \frac{d^2}{2r}$, is valid to calculate h .
 - How many metres is the astronaut above the surface of Mars?

QUESTION 7. [Start a new page]

- (a) Triangle ABC has area equal to 5 cm^2 . If $AC = 4 \text{ cm}$ and $BC = 5 \text{ cm}$, find $\angle ACB$.
- (b) A mining company started three mining towns A, B, and C with a population of 500 each and it was planned that they should grow by roughly 50 inhabitants each, each year for the first 10 years. Only town A grew as planned!
- (i) Write down an expression for the intended population P of the town A, t years after its opening ($t \leq 10$).
- (ii) For various reasons, towns B and C did not grow as planned. Their populations are better modelled by:

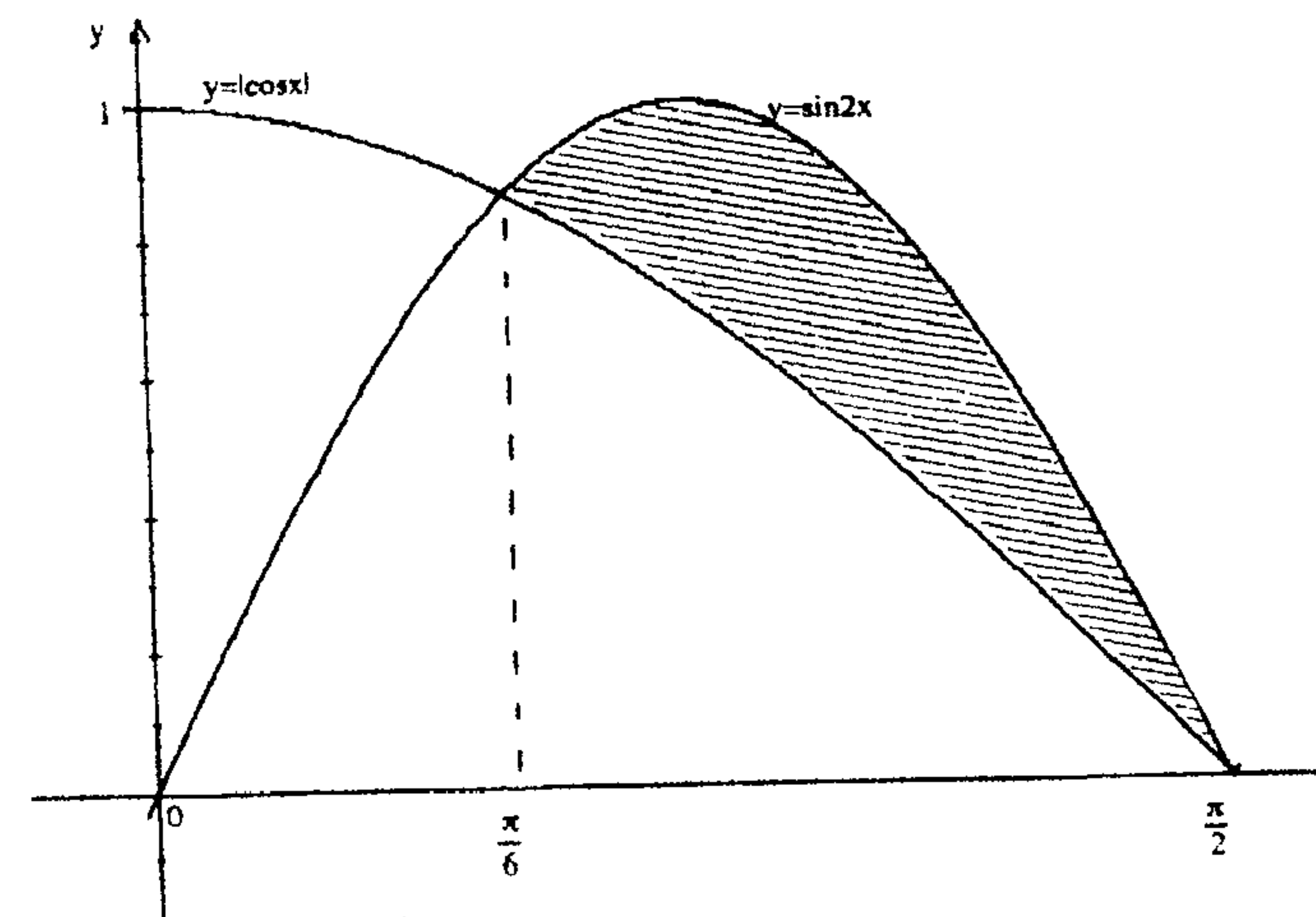
Town B: $\frac{dP}{dt} = -0.3P$

Town C: $P = 100 \left(5 + t - \frac{1}{4}t^2 \right)$

- (1) Prove that the population for town B is given by $P = 500e^{-0.3t}$ after t years.
- (2) Calculate the population of town B after 7 years.
- (3) Find the rate of change of the population of town C after 1, 2, and 3 years.
- (4) What was the maximum population of town C? And in which year was it reached?
- (5) The mining company considers the mines unprofitable and will close the towns if the population gets to below 50. Show that town C will be abandoned before town B.

QUESTION 8. [Start a new page]

- (a) Graph the solution of $2 - x \geq 0$ on the number line.
- (b) Differentiate $f(x) = \frac{x^2}{\ln x}$ and find $f'(e^2)$.
- (c) The diagram below shows the graphs of $y = |\cos x|$ and $y = \sin 2x$ between 0 and $\frac{\pi}{2}$. The area completely enclosed by the two curves is shaded. Show that the shaded area is equal to $\frac{1}{4}$ square units.

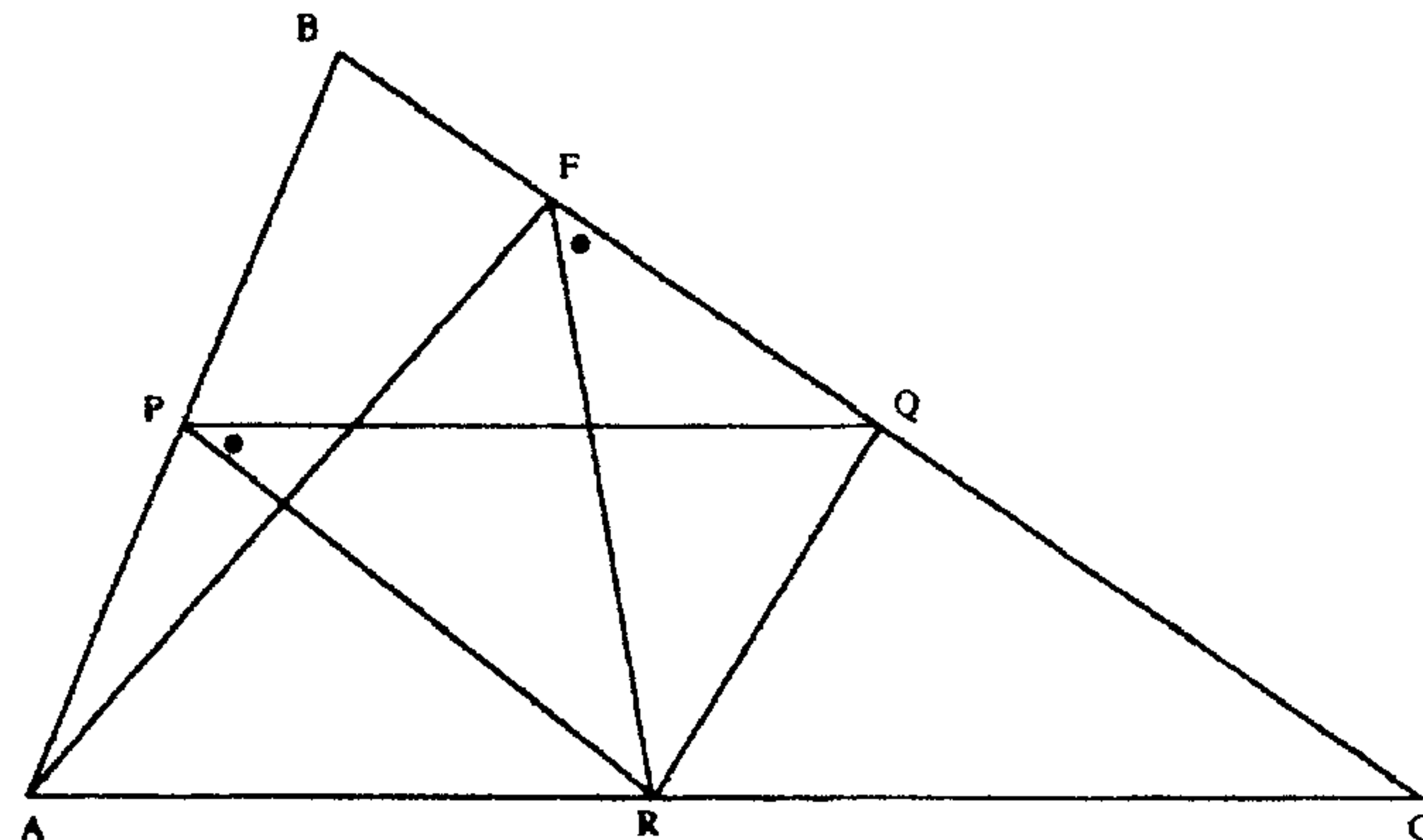


- (d) (i) Copy the diagram of (c) and extend it to $x = 2\pi$, that is, graph $y = |\cos x|$ and $y = \sin 2x$ between 0 and 2π .
- (ii) Find the area of the 3 regions now enclosed by the two curves for $\frac{\pi}{6} \leq x \leq \frac{3\pi}{2}$.

QUESTION 9. [Start a new page]

- (a) (i) Graph $y = \frac{2x}{x^2 + 1}$.
 (ii) Find the area under the curve, the x axis and the lines $x = -1$ and $x = 2$.

- (b) In $\triangle ABC$, P, Q, and R are mid-points of sides AB, BC, and CA, respectively. F is a point in side BC, such that $\angle RPQ = \angle RFQ$.

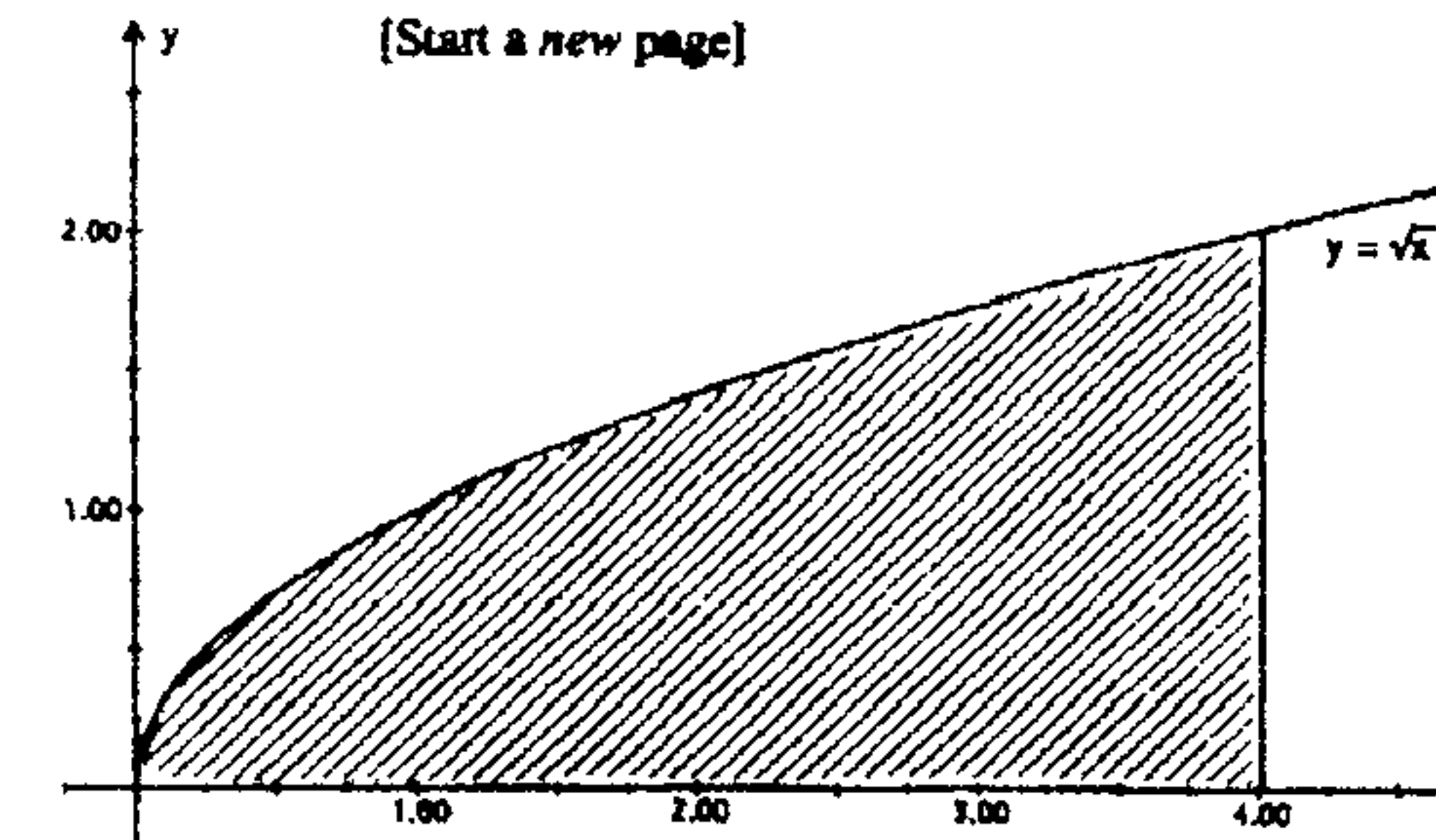


Show that:

- (i) $AC = 2PQ$ (give reasons)
- (ii) $\triangle ABC$ is similar to $\triangle QRP$ (give reasons)
- (iii) $\angle ACB = \angle RFQ$ (give reasons)
- (iv) $\angle RFA = \angle FAR$ (give reasons)
- (v) line AF is perpendicular to line FC (give reasons).

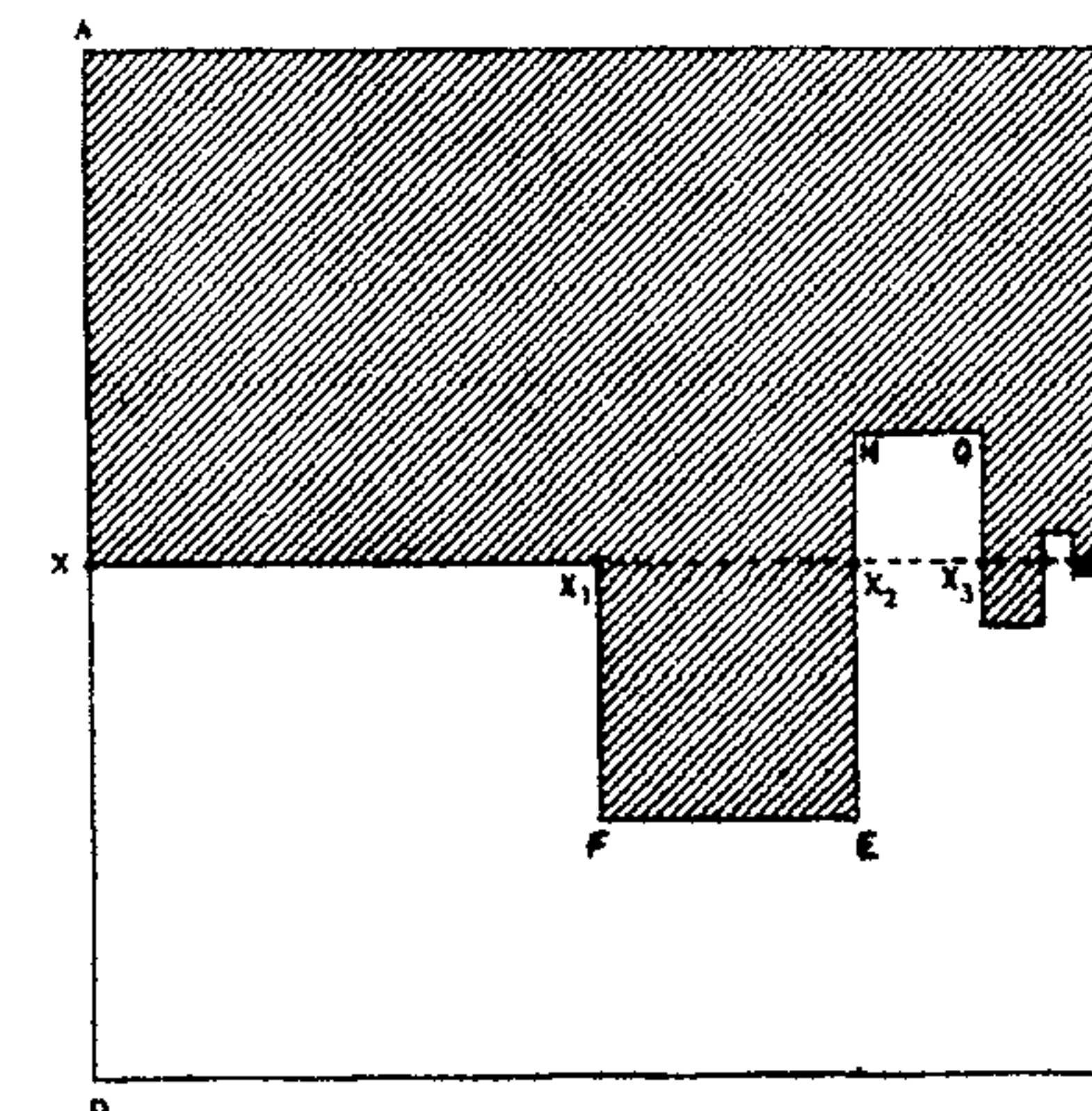
QUESTION 10. [Start a new page]

(a)



The shaded region is bounded by the lines $y = 0$, $x = 4$, and $y = \sqrt{x}$.
 Find the volume of the solid of revolution when the shaded region is rotated about the y -axis.

(b)



$[ABCD]$ is a square of side a . X and Y are mid points of AD and BC respectively.
 X_1 is the mid point of XY and X_1X_2EF is a square.
 X_2 is the mid point of X_1Y and X_2X_3GH is also a square.
 X_3 is the mid point of X_2Y

This process is continued indefinitely, with squares smaller and smaller alternately above and below the line XY.

- (i) Express the area of square X_1X_2EF as a function of a , the side of the square ABCD.
- (ii) Calculate the shaded area as a fraction of the area of the square ABCD.

- (c) Given the relation $\frac{y}{x} + \frac{x+2}{y+1} = 2$, $x \neq 0$, $y \neq -1$

- (i) Show that $y^2 + y(1 - 2x) + x^2 = 0$.
- (ii) Find the greatest integer x for which y is real and rational.

QUESTION 1

- (a) 3.8
 (b) \$50
 (c) $f(1) = \frac{1}{2}$
 $f(2) = \frac{1}{9}$
 (ii) $\frac{1}{18}$
 (d) $-\frac{1}{4}$
 (e) $(a+b)(a-b+2)$

QUESTION 2

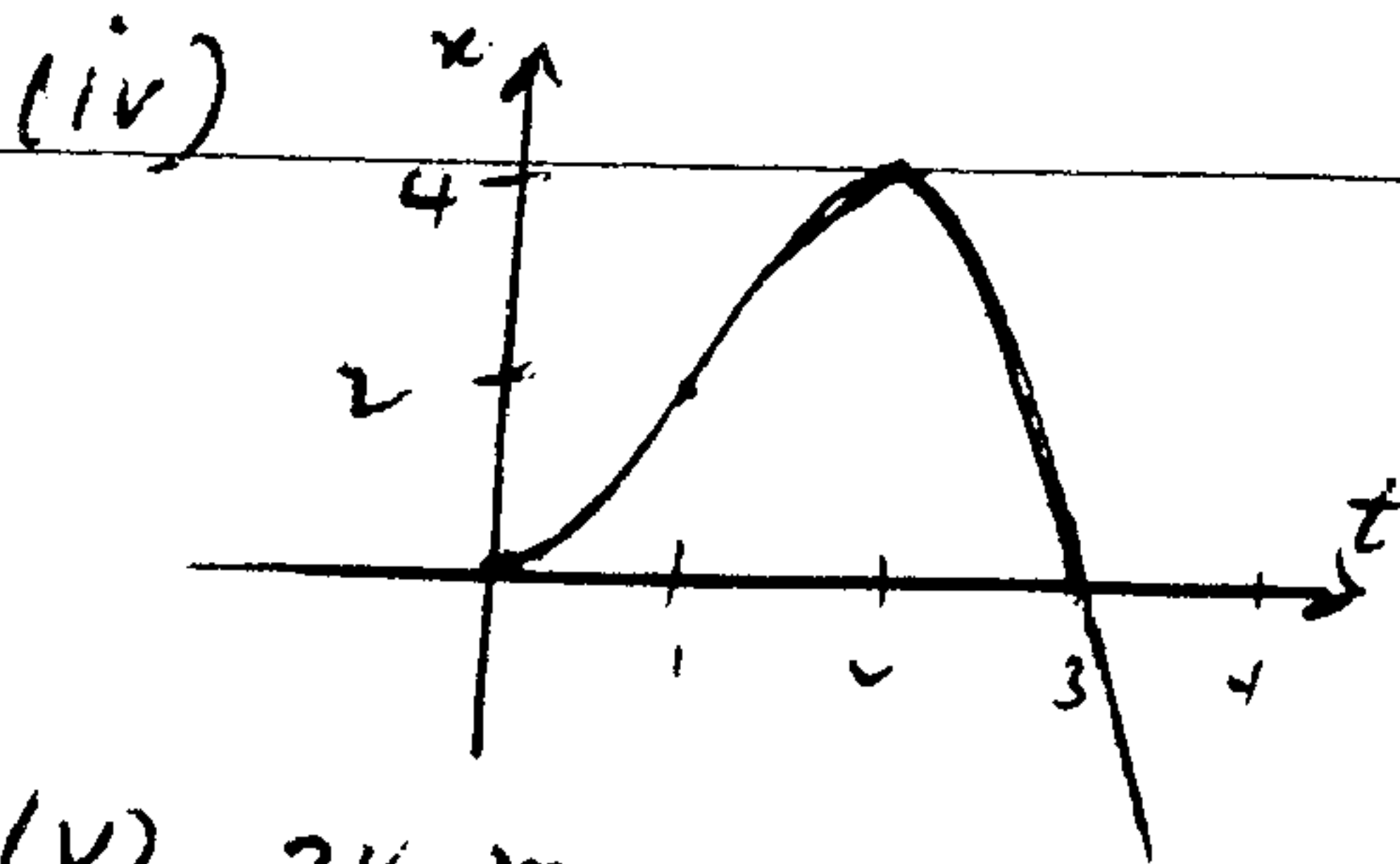
- (a) $x = 7$
 (b) (i) $x = -8, 2$
 (ii) $x \leq -8, x \geq 2$
 (c) (i) (1) $\frac{2}{7}$
 (2) $\frac{4}{7}$
 (ii) $\frac{6}{17}$

QUESTION 3

- (a) (i) 60°
 (ii) —
 (iii) $32\sqrt{3} \text{ cm}^2$
 (b) (i) $(x+1)^2 = 4(\frac{1}{4})(y+6)$
 (ii) $V(-1, -6)$
 $S(-1, -5\frac{3}{4})$
 (iii) $\frac{4}{7}$

QUESTION 4

- (i) $v = 0 \text{ m/s}$
 $a = 6 \text{ m/s}^2$
 (ii) min tp (0,0)
 max tp (2,4)
 (iii) $t = 1 \text{ sec.}$



- (v) 24 m
 (vi) —

QUESTION 5

- (a) (i) $d = 2$
 (ii) $S_{150} = 22500$
 (b) (i) $p = 16, q = -2$
 (ii) —
 (iii) $y = 4x - 16$
 (iv) $T(3, -4)$
 (v) midpt of TM $(3, 2\frac{1}{4})$
 lies on parabola.

QUESTION 6

- (a) $\frac{25\pi}{8} \text{ cm}^2$
 (b) (i) —
 (ii) —
 (iii) —
 (iv) (1) —
 (2) $\div 59 \text{ metres}$

QUESTION 7

- (a) $30^\circ \text{ or } 150^\circ$
 (b) (i) $P = 500 + 50t$
 (ii) (1) —
 (2) $\div 61 \text{ people}$
 (3) 50, 0, -50
 (4) 600

- (5) when $t = 7$,
 population of B $\div 61$
 population of C < 0
 $\therefore$ C closes before B

QUESTION 8

- (a) 
 (b) $f'(x) = \frac{2x \ln x - x}{(\ln x)^2}$
 $f'(e^2) = \frac{3e^2}{4}$
 (c) $\frac{1}{4}$
 (d) (i) —
 (ii) $2\frac{3}{4}$

QUESTION 9

- (a) (i) —
 (ii) $\ln 10$
 (b) —

QUESTION 10

- (a) $128\pi/5$
 (b) (i) $\frac{1}{6}a^2$
 (ii) $\frac{1}{20}$
 (c) (i) —
 (ii) $x = -2$